

Ansh Agrawal

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Education

Manipal University Jaipur – B.Tech, Computer Science and Engineering Aug 2023 – Jun 2027
CGPA: 7.89/10 | Coursework: Data Structures & Algorithms, OOP, DBMS, OS, Computer Architecture

DAV Public School, Sec-49, Gurugram (CBSE) Class XII: 85.2% (2023) | Class X: 95.1% (2021)

Experience

Apollo Tyres Ltd. – Intern, IT & Digital May 2026 – Jul 2026, Gurgaon

- Built a production **AI agent on AWS** that converts natural-language questions into **openCypher** queries over **Amazon Neptune** property graphs, grounding generation in schema docs indexed via **Bedrock Knowledge Base + S3** (RAG).
- Designed a **hybrid query router**: RAG-grounded prompts for known query patterns, dynamic generation for complex multi-hop traversals across entity relationships.
- Deployed the service on **AWS AgentCore Runtime** (Strands Agents SDK) with cross-session memory; containerized for Linux ARM64 and exposed over a live HTTPS endpoint.

Projects

Study Buddy – Voice-Based Agentic Tutor (3rd Place, Assesli Hackathon) [GitHub](#)

Next.js, TypeScript, Supabase (Postgres, Edge Functions, Storage), AWS Lambda, Gemini, Gemini TTS

- Built a 4-phase tutoring agent (Analyze, Teach, Assess, Recommend) that adapts difficulty to learner responses, with voice input/output via Web Speech API and Gemini TTS.
- Re-architected from a tunneled Node server into an **event-driven serverless pipeline**: DB inserts trigger a Supabase Edge Function that invokes **AWS Lambda** for AI processing, with generated audio stored in Supabase Storage and pushed to clients via real-time updates.

Smart Inventory – Inventory Management SaaS for SMBs [GitHub](#) | [Live](#)

Next.js 14, TypeScript, PostgreSQL (Supabase), Vercel Cron, Tesseract.js, Gemini Vision, Groq

- Guaranteed stock consistency with a **ledger-based inventory model** (PostgreSQL triggers) and **atomic recipe-based deductions** executed as single-transaction RPCs.
- Automated stock entry via **OCR bill scanning** (Tesseract.js + Gemini Vision); deployed to production on Vercel.
- Scheduled 3 background agents (reorder, shrinkage, demand forecasting) via Vercel Cron with human approval; secured user API keys with **AES-256-GCM** encryption (BYOK).

ReconAI – Payment Reconciliation Engine (Razorpay Buildathon) [GitHub](#) | [Blog](#)

Python, FastAPI, PostgreSQL (Supabase), LangGraph, Next.js, TypeScript

- Built a 5-stage pipeline reconciling orders against bank settlements – **deterministic matching first** (fee deductions, rounding, missing IDs), LLM fallback only for ambiguous cases – achieving **100% precision and recall** on hidden ground truth.
- Diagnosed a **race condition** causing duplicate settlement claims; redesigned as a two-pass batch pipeline with **DB-level claim locking**, making runs crash-safe.
- Shipped a FastAPI backend and Next.js review dashboard where every manual override requires explicit confirmation.

Technical Skills

Languages: C++, Python, Java, SQL, C, TypeScript, JavaScript, Kotlin

Backend & Web: FastAPI, Node.js, Next.js, Flask, REST APIs

Databases: PostgreSQL, Supabase, MongoDB, Firebase

Cloud & Tools: AWS (Lambda, S3, Neptune, Bedrock, AgentCore), Supabase Edge Functions, Vercel, Linux, Git

AI/LLM: LangGraph, LangChain, RAG, ChromaDB, FAISS, Gemini, Groq

Achievements

- Amazon ML Challenge 2026** – Top 13% of 8,300+ teams as Team Lead (team of 4); built a business entity-resolution pipeline across 3 data sources (macro F0.5: 0.98)
- 1st Place** – The Hackathon, Manipal University Jaipur (Apr 2025)
- 4th Place** – CODESTORM, BITS Goa (Feb 2025); **4th Place** – ML Hackathon, IIT Kanpur Techkriti'25 (Mar 2025)
- DSA:** Completed NeetCode 150 and LeetCode SQL 50